

Macroeconomics

The Goods Market

(Blanchard, Macroeconomics 8th edition Ch. 3)

Short Run, Medium Run, and Long Run

- ▶ *Short run* (e.g., a few years): year-to-year movements in output are primarily driven by movements in demand.
- ▶ *Medium run* (e.g., a decade): the economy tends to return to the level of output determined by supply factors, such as the capital stock, the level of technology, and the size of the labor force.
- ▶ *Long run* (e.g., a few decades or more): the economy depends on its ability to innovate and introduce new technologies, and how much people save, the quality of the education system, government, etc.

Keynesian Economics

- ▶ John Maynard Keynes: influential British economist, especially during Great Depression
- ▶ Focused on demand-side theory and changes in the economy over the short run
- ▶ “In the long run we all dead.”



The Goods Market

- ▶ When economists think about year-to-year movements in economic activity, they focus on the interactions among production, income, and demand:
 - ▶ Δ demand for goods $\rightarrow \Delta$ production
 - ▶ Δ production $\rightarrow \Delta$ income
 - ▶ Δ income $\rightarrow \Delta$ demand for goods
- ▶ Demand, production, and income are all interconnected!

Definitions of GDP

1. GDP is the value of final goods and services produced in the economy during a given period.
 2. GDP is the sum of value added in the economy during a given period.
 3. GDP is the sum of incomes in the economy during a given period.
- ▶ **Nominal GDP** is the sum of the quantities of final goods produced times their current price.
 - ▶ Nominal GDP increases for two reasons:
 - ▶ The production of most goods increases.
 - ▶ The price of most goods increases.
 - ▶ **Real GDP** is the sum of quantities of final goods times constant (not current) prices.

GDP vs GNP

- ▶ Measure of economic activity
- ▶ Careful to distinguish between what is produced inside and outside the economy
- ▶ Gross domestic product (GDP): a measure of the value of all the goods and services newly produced in an economy during a specified period of time
- ▶ GDP measures the volume of production *within* a country's borders
- ▶ Gross national product (GNP): a measure of the value of all the goods and services produced by *the country's factors of production* during a specified period of time
- ▶ Factors of production: labor, capital, natural resources, etc.
- ▶ GNP measures the volume of production by a country's factors of production, regardless of whether it is within the country's borders

The Composition of GDP

- ▶ **Consumption (C)**
 - ▶ Goods and services purchased by consumers
 - ▶ Largest component
- ▶ **Investment (I) or fixed investment**
 - ▶ Sum of nonresidential investment (plants, machines) and residential investment
 - ▶ Both generate output or services in the future
 - ▶ Most volatile
- ▶ **Government spending (G)**
 - ▶ Purchases of goods and services by federal, state, and local governments (airplanes, office equipment, government employees)
 - ▶ Excludes government transfers

Denmark abolishes public holiday to boost defense spending

02/28/2023

The government has argued that by getting rid of "Great Prayer Day" as a public holiday, more funds will be generated to flow into public spending and boost defense. Thousands have protested the move.



<https://www.youtube.com/watch?v=8hKWf1wVT6I&t=37s>

The Composition of GDP (con.)

- ▶ **Exports (X)**

- ▶ Purchases of domestic goods and services by foreigners

- ▶ **Imports (IM)**

- ▶ Purchases of foreign goods and services by domestic consumers, firms and government
- ▶ Net exports or trade balance: $X - IM$
- ▶ Trade surplus $\Leftrightarrow X > IM$
- ▶ Trade deficit $\Leftrightarrow X < IM$

- ▶ **Inventory investment**

- ▶ Difference between production and sales
- ▶ e.g., production $>$ sales $\Rightarrow$ positive

The Composition of GDP: US (2018)

		Billions of Dollars	Percent of GDP
	GDP (Y)	20,500	100.0
1	Consumption (C)	13,951	68.0
	Investment (I)	3,595	17.5
2	Nonresidential	2,800	13.6
	Residential	795	3.8
3	Government spending (G)	3,522	17.2
	Net exports	-625	-3.0
4	Exports (X)	2,550	12.4
	Imports (IM)	-3,156	-15.4
5	Inventory investment	56	0.2

Source: Survey of Current Business, February 2019, Table 1-1-5

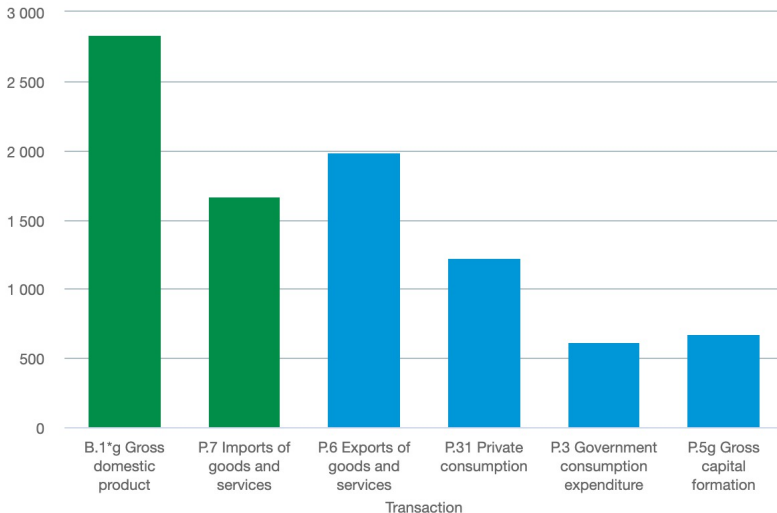
Danish GDP

- ▶ <https://www.dst.dk/en/Statistik/emner/oekonomi/nationalregnskab>
- ▶ <https://www.dst.dk/en/Statistik/emner/oekonomi/nationalregnskab/noegletal-for-nationalregnskabet-bnp>
- ▶ “The aim of the national accounts is to provide an overall picture of the economy. The National Accounts system is a logical and coherent set of definitions and classifications, without which it would not be possible to obtain an overview of the immense number of financial transactions that take place in the economy during a period. ...”

Danish GDP and its components

The balance of demand and supplys main components, billion DKK

Price unit: Current prices, (bill. DKK.) | Time: 2022:



The Demand for Goods

- ▶ This identity defines the total demand for goods (Z):

$$Z \equiv C + I + G + X - IM$$

- ▶ In a closed economy ($X = IM = 0$):

$$Z \equiv C + I + G$$

- ▶ Implicitly assumes that firms produce the same good which can then be used by consumers for consumption, by firms for investment, or by the government.
- ▶ Also, assume that for a given price level, firms are willing to supply any amount of the good, i.e., we focus on the demand side.

Consumption

- ▶ Consumption (C) is a function of disposable income (Y_D): income that remains once consumers have received government transfers and paid their taxes.

$$\oplus$$
$$C = C(Y_D)$$

- ▶ $C(Y_D)$ is called the consumption function: behavioral equation that captures the behavior of consumers, e.g., linear:

$$C = c_0 + c_1 Y_D$$

- ▶ $c_1 < 1$: (marginal) propensity to consume (MPC); c_0 is what people would consume if their disposable income equals zero (e.g., dissave: sell assets or borrow)
- ▶ Changes in c_0 reflect changes in consumption for a given level of disposable income (e.g., easier to borrow, optimistic about future).

Consumption (con.)

Consumption (con.)

- ▶ Disposable income is: $Y_D \equiv Y - T$ where Y is income and T is taxes minus government transfers.
- ▶ Replacing Y_D gives:

$$C = c_0 + c_1(Y - T)$$

- ▶ Higher income increases consumption, but less than one for one.
- ▶ Higher taxes decrease consumption, also less than one for one.

East Asia

With China's Gen Z consumers craving value over status, luxury brands and global markets brace for impact

Amid slower economic growth, young consumers in China are increasingly prioritising value and suitability over pursuing status symbols. This change in attitude is affecting luxury brands and global markets.



Bong Xin Ying

29 Aug 2024 06:00AM

(Updated: 29 Aug 2024 06:04PM)

[https://www.channelnewsasia.com/east-asia/
china-gen-z-rational-spending-budget-economy-4570191](https://www.channelnewsasia.com/east-asia/china-gen-z-rational-spending-budget-economy-4570191)

Consumption in China

- ▶ a new generation of frugal Gen Z consumers in China, ... who experts say have been greatly reevaluating their spending habits and purchases, becoming more rational and discerning with their choices as they grapple with a multitude of issues like high youth unemployment rates amid growing economic uncertainty.
- ▶ “About 80 per cent of my purchases are made online,” Ms Mi said. “The price advantage determines (where) I make my purchase.”
- ▶ This rapid shift towards slower, more rational consumption and practicality - is a sign of a maturing economy, analysts said, and also signals an important transition for China from its booming years, when luxury and status were prioritised.
- ▶ Mr Yu noted that “the government is very clear that they do want to drive domestic consumption, and today’s situation is not ideal”

'Little luxuries': The affordable treats driving consumer spending

4 January 2024

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Tatiana Walk-Morris
Features correspondent

- ▶ [https://www.bbc.com/worklife/article/20240104-little-luxuries-the-affordable-treats-driving-con](https://www.bbc.com/worklife/article/20240104-little-luxuries-the-affordable-treats-driving-consumer-spending)
- ▶ Consumers may be easing up on splurge purchases, but they're still buying pick-me-up products that make them feel special.
- ▶ This desire for these "affordable luxuries" is common in difficult economic times. Some economists refer to the phenomenon as the "lipstick index": a small economic bump led by budget-conscious consumers seeking out relatively affordable splurges, like small cosmetics and perfumes.
- ▶ "The 'lipstick effect' of spending on small treats during more challenging economic times is real, and we have seen it in action this year," says Mercer. "it encompasses more than beauty and includes categories such as treat foods."

Taxes

- ▶ Note that we have specified a **lump-sum** tax T
- ▶ Every household/consumer faces the same tax, i.e., this is not a tax *rate*
- ▶ In reality, taxes are **distortionary**
- ▶ Distortionary taxes change incentives depending on household income Y
- ▶ We work with lump-sum taxes for simplicity in (most of?) this course

Investment

- ▶ *Endogenous* variables: variables that depend on other variables in the model
- ▶ *Exogenous* variables: variables not explained within the model but are instead taken as given

$$I = \bar{I}$$

- ▶ A bar on investment means investment is taken as given.
- ▶ Because investment changes output in the future, we would need a two-period model to fully describe the investment decision.

Government expenditures

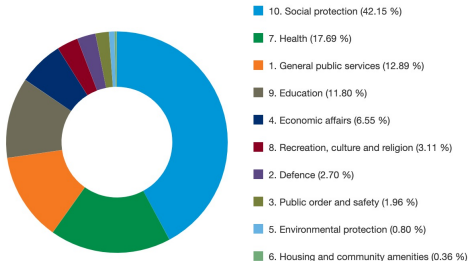
- ▶ T and G describe fiscal policy-the choice of taxes and spending by the government.
- ▶ G and T are exogenous because:
 - ▶ Governments do not behave with the same regularity as consumer or firms.
 - ▶ We will typically treat G and T as variables chosen by the government and will not try to explain them within the model.

Danish government expenditures and taxes

- ▶ <https://www.dst.dk/en/Statistik/emner/oekonomi/offentlig-oekonomi/offentlig-sektors-oekonomi>
- ▶ <https://skat.dk/data.aspx?oid=2244282>

General government expenditure by function (COFOG) (m DKK)

Time: 2022:



Automatic stabilizers

- ▶ <https://www.vox.com/2020/3/9/21167391/coronavirus-economy-stimulus-recession>
- ▶ The right solution is to agree to fiscal stimulus but require it to take the form of creating new automatic stabilizers. Programs that will, in the future, trigger immediately, without any additional congressional action.
- ▶ Last spring, the Brookings Institution... lays out a bunch of automatic stabilizer ideas. One of the most important was developed by Claudia Sahm, a former Federal Reserve staff economist ...

Automatic stabilizers (con.)

- ▶ Her research created something called the Sahm Rule Recession Indicator, which serves as a good trigger point – historically, when the unemployment rate over the past three months rises at least 0.5 percentage points above the average over the past 12 months, a recession is imminent.
- ▶ Sahm's idea is that when the rule triggers, the federal government should send cash directly to every household.
- ▶ That would put the US in position to respond quickly – indeed, immediately – if the unemployment rate does spike.
- ▶ The biggest problem with creating an automatic fiscal stabilizer program, after all, has been that Congress doesn't like to focus on things until there's an emergency.

The Determination of Equilibrium Output

- ▶ Assume $X = IM = 0$, and that firms do not hold inventories:

$$Z = C + I + G$$

- ▶ Replacing C and I :

$$Z = c_0 + c_1(Y - T) + \bar{I} + G$$

- ▶ Equilibrium in the goods markets requires

$$Y = Z$$

- ▶ This is an **equilibrium condition**.
- ▶ Models include three types of equations: identities, behavioral equations, and equilibrium conditions.

The Determination of Equilibrium Output

- ▶ Replacing Z gives

$$Y = c_0 + c_1(Y - T) + \bar{I} + G$$

- ▶ In equilibrium, production (Y) is equal to demand, which in turn depends on income (Y), which is itself equal to production.
- ▶ Macroeconomists always use three tools:
 1. Algebra to make sure that the logic is correct
 2. Graphs to build the intuition
 3. Words to explain the results

The Determination of Equilibrium Output: Algebra

The Determination of Equilibrium Output: Algebra

- ▶ **Autonomous spending:** $[c_0 + \bar{I} + G - c_1T]$
- ▶ Autonomous spending is positive because if $T = G$ (*balanced budget*) and c_1 is between 0 and 1, then $G - c_1T$ is positive.
- ▶ $1/(1 - c_1)$ is the **multiplier**, which is larger when c_1 is closer to 1.
- ▶ e.g., $c_1 = 0.6$, the multiplier =

- ▶ Where does the multiplier effect come from?

The Determination of Equilibrium Output: Graphical

- Steps to characterize the equilibrium graphically:
 1. Plot production as a function of income. Because production equals income, their relation is the 45-degree line.
 2. Plot demand as a function of income.

$$Z = (c_0 + \bar{I} + G - c_1T) + c_1Y$$

3. In equilibrium, production equals demand.

The Determination of Equilibrium Output: Graphical

Example: c_0 increase

The Determination of Equilibrium Output: Graphical

- ▶ AB: first-round increase in demand / production
- ▶ BC: first-round increase in income
- ▶ CD: second-round increase in demand / production
- ▶ DE: second-round increase in income
- ▶ The total increase in production after $n + 1$ rounds:

$$1 + c_1 + c_1^2 + \dots + c_1^n$$

- ▶ which is a geometric series with a limit of $1/(1 - c_1)$.

The Determination of Equilibrium Output: Verbal

- ▶ Production depends on demand, which depends on income, which is itself equal to production.
- ▶ An increase in demand leads to an increase in production and income, which in turn leads to a future increase in demand.
- ▶ The increase in output that is larger than the initial shift in demand, by a factor equal to the multiplier.
- ▶ The multiplier depends on the propensity to consume, which can be estimated using econometrics (set of statistical methods).
- ▶ Adjustment of output over time: **dynamics** of adjustment.
- ▶ How long the adjustment takes depends on how and when firms revise their production schedule.

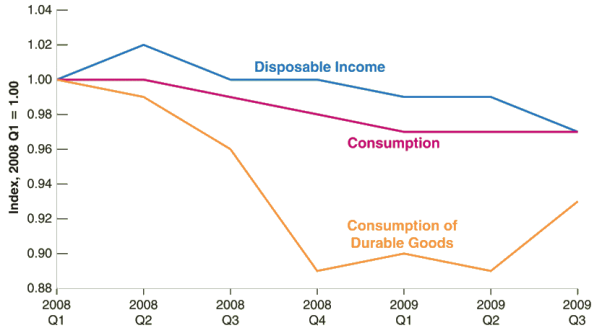
Fiscal Multiplier

- ▶ <https://cepr.org/voxeu/columns/economic-effects-military-support-ukraine-evidence-fiscal->
- ▶ “While it appears that the multiplier did not change materially following the first invasion of Ukraine in 2014 or in 2022, in the short term, an expansion of military spending due to the war in Ukraine may have a stimulative effect on output in donor countries. The economic cost of military assistance to Ukraine may be materially smaller when this domestic output effect is taken into account.”
- ▶ “We estimate that a \$1 increase in military spending by the countries that supported Ukraine in 2022 is associated with a \$0.65 increase in output within a year and a \$0.79–\$0.87 increase over the following two years. The cumulative multipliers decline at longer horizons but remain positive and significant for five years.”

Global Financial Crisis

- ▶ Why would consumers decrease consumption if their disposable income has not changed?
- ▶ i.e., why might c_0 decrease, leading in turn to a decrease in demand, output, and so on?
- ▶ May start worrying about the future and decide to save more.
- ▶ This is precisely what happened at the start of the crisis.

Global Financial Crisis (con.)



Source: Calculated using series DPIC96, PCECC96, PCDGCC96: Federal Reserve Economic Data (FRED)
<http://research.stlouisfed.org/fred2/>

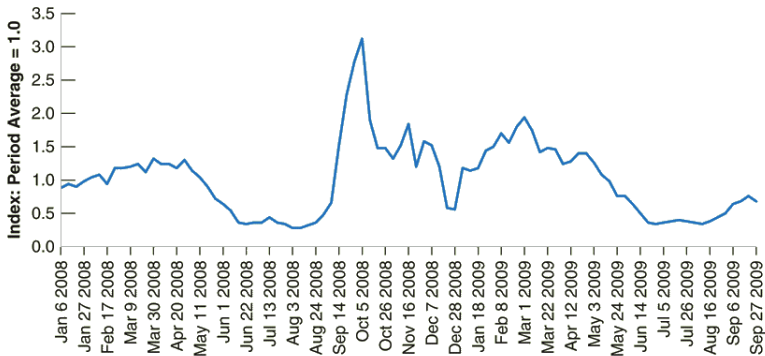
Global Financial Crisis (con.)

- ▶ Despite large fall in GDP, disposable income did not initially move much: even increased in 2008 Q1.
- ▶ But consumption was unchanged from 2008 Q1-Q2 and then fell before disposable income fell: 3% in 2009 relative to 2008, more than the decrease in disposable income.
- ▶ During 2008 Q3-Q4, consumption of durables dropped sharply: down 10% in Q4 relative to Q1
- ▶ Psychological fallout of financial crisis: on September 15, 2008, Lehman Brothers went bankrupt $\Rightarrow$ appeared that financial system might collapse.

Global Financial Crisis (con.)

- ▶ For most people, the main sign of trouble was what they read in newspapers.
- ▶ Even though they still had their job and received their monthly income checks, the events reminded them of the stories of the Great Depression and the pain that came with it.
- ▶ Given the uncertainty about what was going on, delay the purchases you could afford to delay, e.g., new car or TV.
- ▶ One way to see this is to look at the Google Trends series that gives the number of searches for “Great Depression” .

Global Financial Crisis (con.)



Source: Google Trends, "Great Depression."

Goods-Market Equilibrium: Alternative Model

- ▶ John Maynard Keynes articulated an alternative model that focuses instead on investment and saving in the General Theory of Employment, Interest and Money in 1936.
- ▶ Private saving (S) is

$$\begin{aligned} S &\equiv Y_D - C \\ &= Y - T - C \end{aligned}$$

- ▶ By definition, public saving $= T - G$.
- ▶ Public saving $> 0 \Leftrightarrow$ Budget surplus
- ▶ Public saving $< 0 \Leftrightarrow$ Budget deficit

IS Relation

Equivalence

- Equivalent ways to state condition for equilibrium in goods market:

$$\text{Production} = \text{Demand}$$

$$\text{Investment} = \text{Saving}$$

The Paradox of Saving

- ▶ We are told about the virtues of thrift as we grow up, but our model tells a different story.
- ▶ Suppose that consumers decide to save more, so c_0 decreases, which implies that output decreases
- ▶ Appears to be ambiguous effect on savings:

$$S = -c_0 + (1 - c_1)(Y - T)$$

- ▶ Actually, savings cannot change either, because in equilibrium:

$$I = S + (T - G)$$

- ▶ Although people want to save more at a given level of income, their income decreases by an amount such that their saving is unchanged.

The Paradox of Saving (con.)

- ▶ So should the government tell people to be less thrifty?
- ▶ Not necessarily. The results of this simple model are of much relevance in the short run.
- ▶ An increase in the saving rate is likely to lead over time to higher saving and higher income.
- ▶ A warning remains, however: Policies that encourage saving might be good in the medium run and in the long run, but they can lead to a reduction in demand and in output, and perhaps even a recession, in the short run.

Is the Government Omnipotent? A Warning

- ▶ Results imply that the government can choose the level of G or T to affect the level of output it wants.
- ▶ However, there are many aspects of reality that we have not incorporated in our model:
 - ▶ Changing G or T is not easy.
 - ▶ Investment and imports may change, making it hard for governments to assess the effects of their policies.
 - ▶ Expectations are likely to matter.
 - ▶ The effects on output may be unsustainable in the medium run.
 - ▶ Cutting T or increasing G can lead to large budget deficits and public debt in the long run.